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SEAT No. :

PE-544

[Total No. of Pages : 2

[6578]-17

**S.E. (Electronics and Computer Engineering) (Insem.)
DATA STRUCTURES AND ALGORITHMS
(2019 Pattern) (Semester - III) (204184)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2 & Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

Q1) a) What is pointer? What are the advantages using pointer? Explain pointer declaration and its initialization with an example. **[5]**

b) Explain the following **[5]**

- i) Call by value
- ii) Call by reference

c) Write a C function with and without pointers to arrays for checking whether the given string is a Palindrome or not. **[5]**

OR

Q2) a) Explain constants, variable and keywords in C with example. **[5]**

b) What is union? What is difference between structure and union? **[5]**

c) Write the syntax for opening a file with various modes and closing a file. **[5]**

Q3) a) Sort the following numbers 38,27,43,3,9,82,10 using Bubble sort. **[5]**

b) Explain Binary Search algorithm with example. **[5]**

c) What is time complexity of algorithm? Write down time complexity of each searching and sorting algorithm. **[5]**

P.T.O.

OR

Q4) a) What is a difference between internal sorting and external sorting? Sort the following numbers using selection sort [5]

25,17,31,13,2

b) Sort the following data using merge sort and selection sort. [5]

142, 317, 45, 222, 187

c) Compare linear search and Binary search. Write algorithm to search the element in list using linear search. [5]

